# "Up and Running" With the Bridge2AI-Voice Dataset

**Voice AI Symposium Hackathon**, May 2026

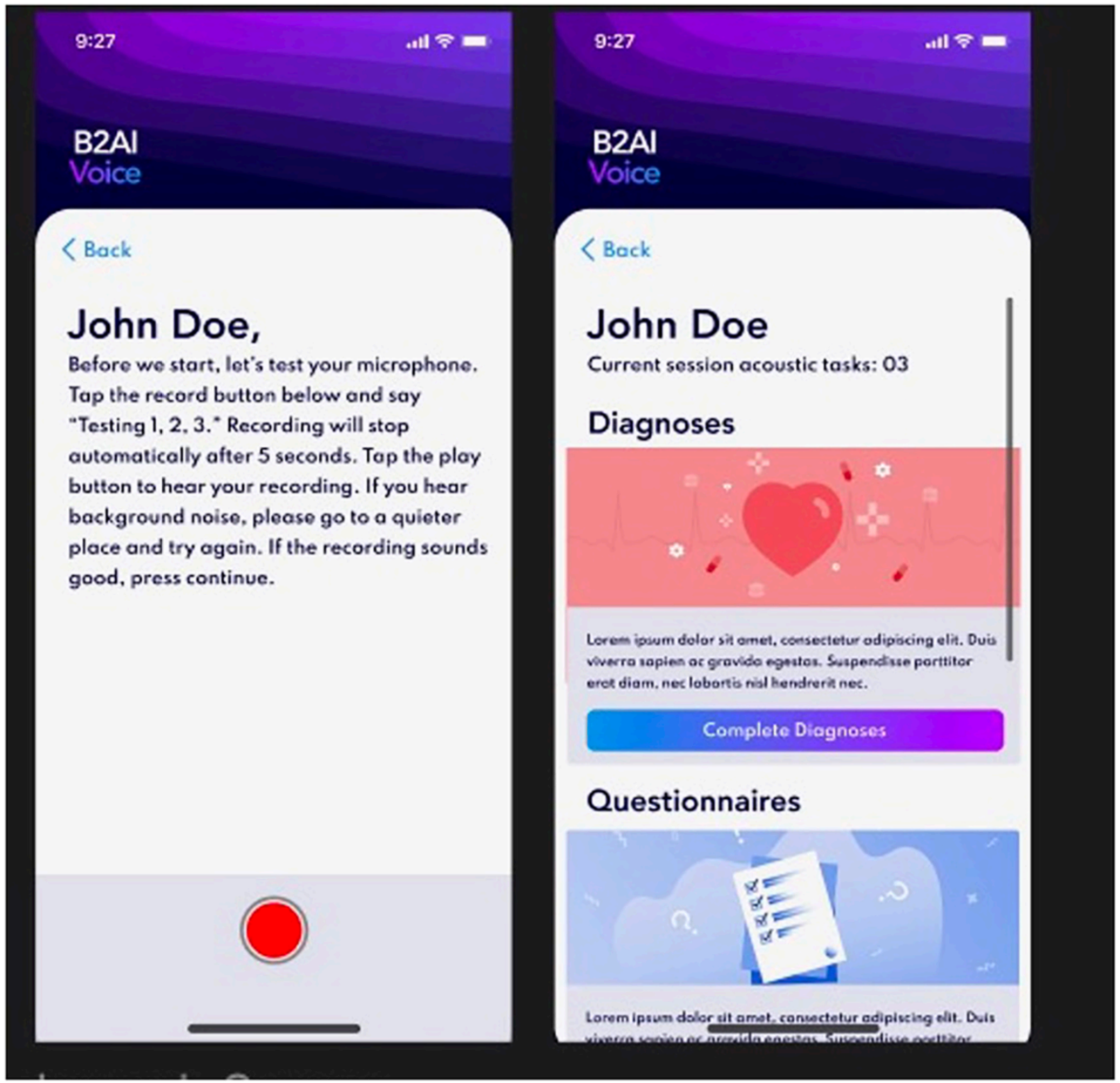
Steven Bedrick, PhD
Oregon Health & Science University

# Today's Plan

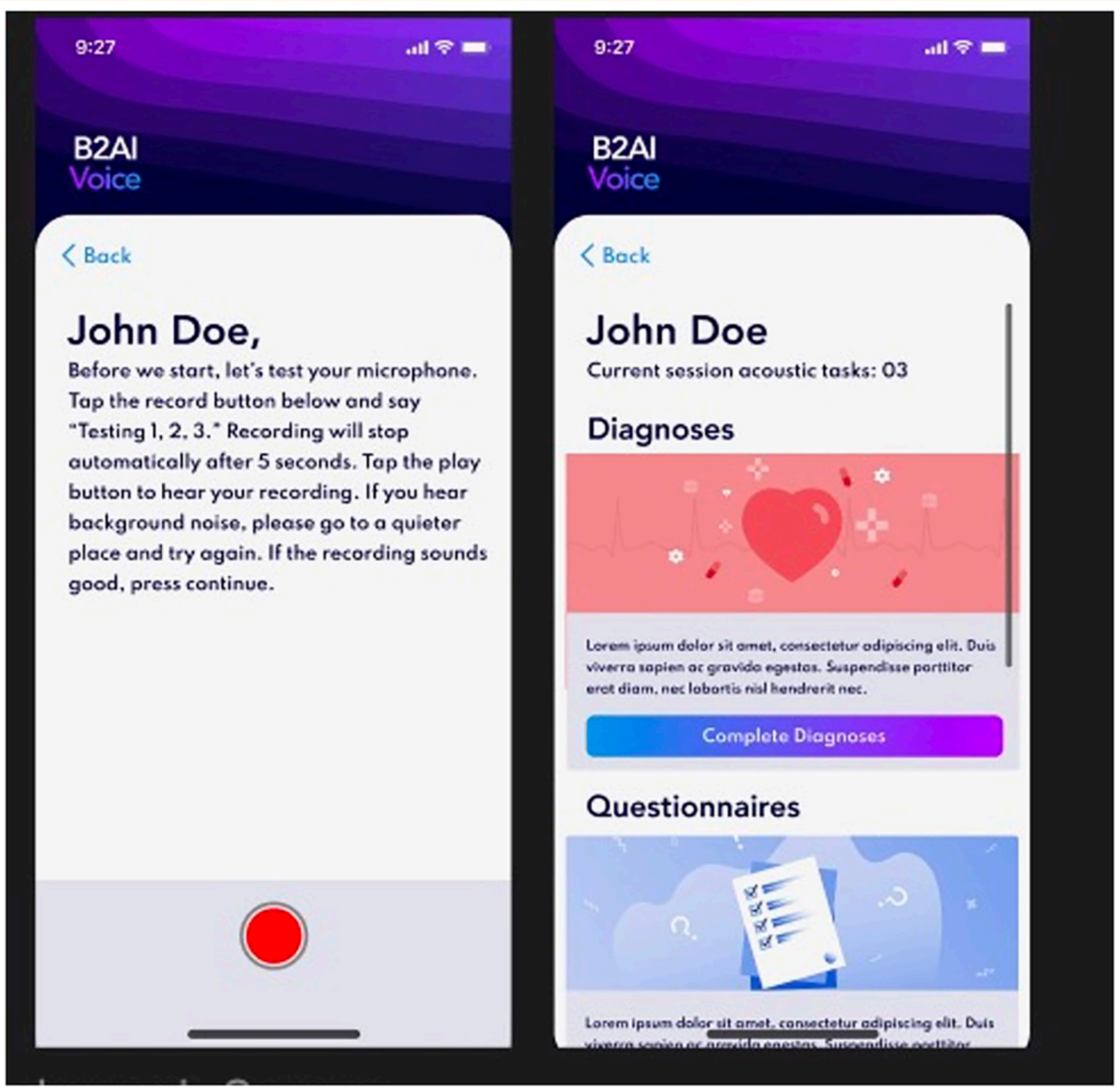
**Our dataset from 30,000 feet**

Getting hands dirty: problem formulation with our data

Practical demonstrations & exercises




Moothedan E, Boyer M, Watts S, Abdel-Aty Y, Ghosh S, Rameau A, et al. The Bridge2AI-voice application: initial feasibility study of voice data acquisition through mobile health. Front Digit Health. 2025 Apr 15;7:1514971.







Moothedan E, Boyer M, Watts S, Abdel-Aty Y, Ghosh S, Rameau A, et al. The Bridge2AI-voice application: initial feasibility study of voice data acquisition through mobile health. Front Digit Health. 2025 Apr 15;7:1514971.

B2Ai Voice Dataset - Adult Da

b2ai-voice.org/bridge2ai-voice-dataset/adult-dashboard/

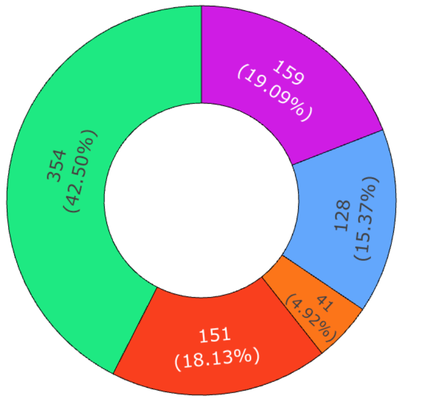
Version: v3.0.0

# Overview

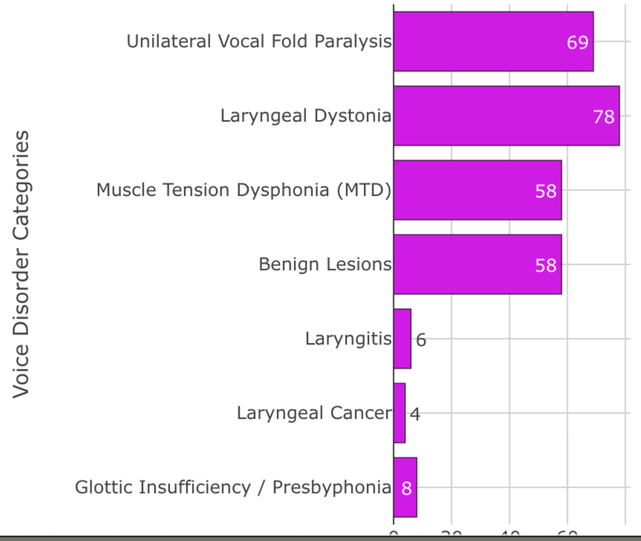
| Number of Participants | Number of Recordings | Total of Questionnaires | Total of Acoustic Tasks | Total Hours of Recordings |
|---|---|---|---|---|
| 833 | 32249 | 9047 | 12042 | 158.52 |

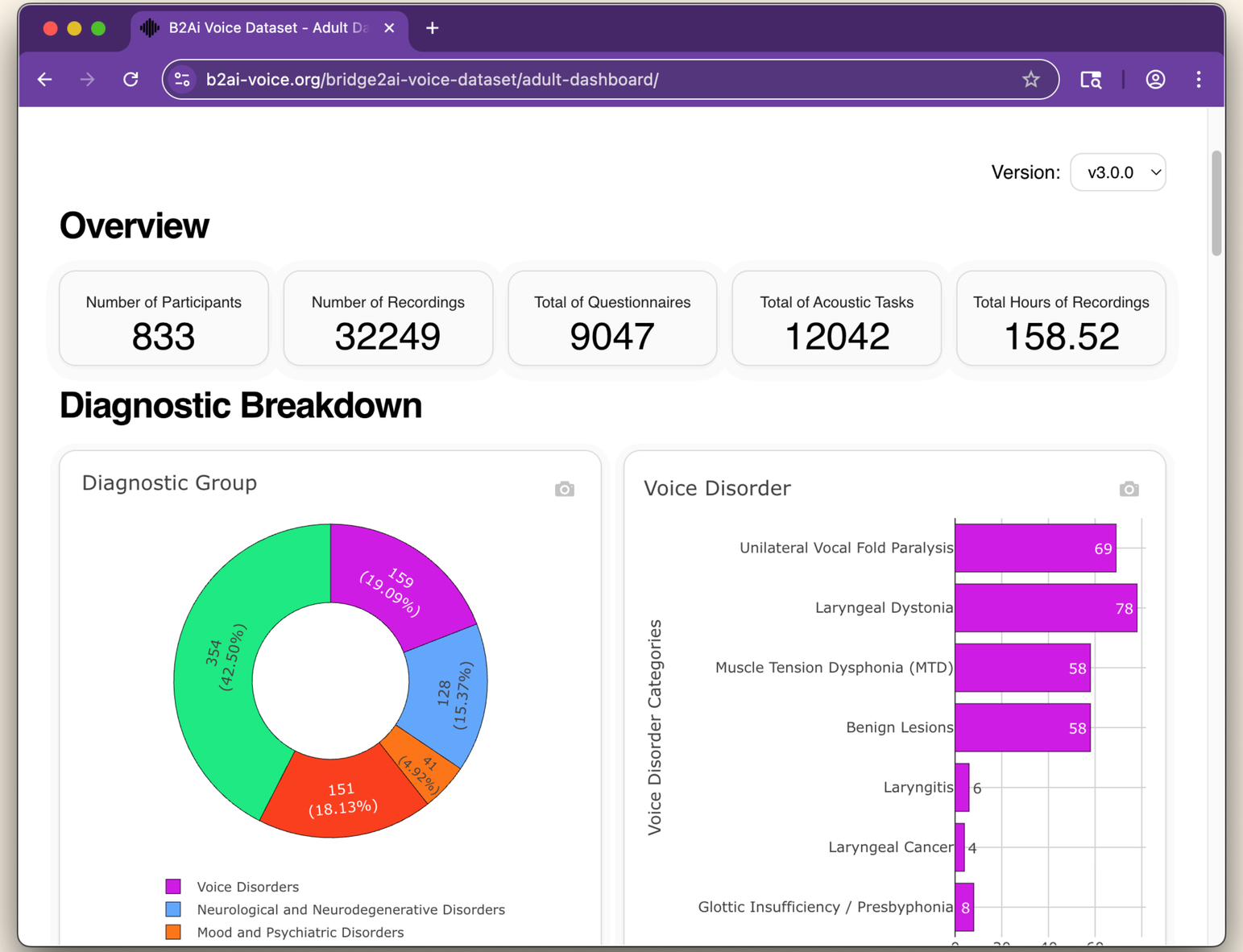
# Diagnostic Breakdown

## Diagnostic Group




## Voice Disorder






https://docs.b2ai-voice.org/

# The data are available in two different releases:

# The data are available in two different releases:

1. Derived features

# The data are available in two different releases:

1. Derived features

    *Credentialed access*

# The data are available in two different releases:

1. Derived features

    *Credentialed access*

    *Questionnaire data, and extracted features from audio recordings for acoustic tasks*

# The data are available in two different releases:

1. Derived features

    *Credentialed access*

    *Questionnaire data, and extracted features from audio recordings for acoustic tasks*

        *... including computed MFCCs and spectrograms.*

**The data are available in two different releases:**

1. Derived features

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   *Questionnaire data, and extracted features from audio recordings for acoustic tasks*

   *... including computed MFCCs and spectrograms.*

2. Raw audio data

**The data are available in two different releases:**

1. Derived features

   *Credentialed access*

   *Questionnaire data, and extracted features from audio recordings for acoustic tasks*

   *... including computed MFCCs and spectrograms.*

2. Raw audio data

   *Institutional DUA required*

**The data are available in two different releases:**

1. Derived features

   *Credentialed access*

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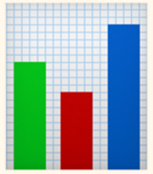
   *... including computed MFCCs and spectrograms.*

2. Raw audio data

   *Institutional DUA required*

   *Includes actual audio data from acoustic tasks*

# Both Releases:



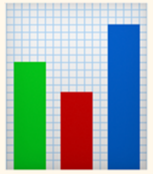




Questionnaire responses, dx status, demographics, confounders, etc.

Low DM, Rao V, Randolph G, Song PC, Ghosh SS. Identifying bias in models that detect vocal fold paralysis from audio recordings using explainable machine learning and clinician ratings. PLOS Digit Health. 2024 May;3(5):e0000516.

**Both Releases:**

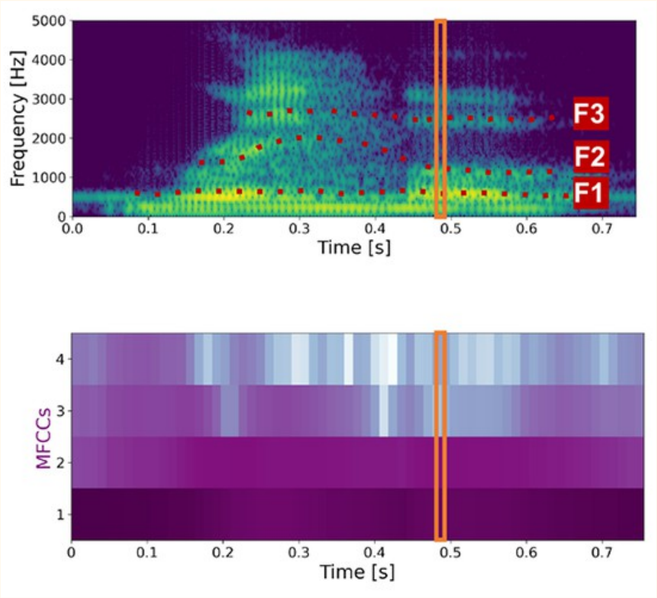






Questionnaire responses, dx status, demographics, confounders, etc.

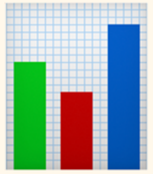
# 1. Derived features




Computed stats: jitter, shimmer, spectral features, transcriptions, loudness, SI-SDR, etc.

Low DM, Rao V, Randolph G, Song PC, Ghosh SS. Identifying bias in models that detect vocal fold paralysis from audio recordings using explainable machine learning and clinician ratings. PLOS Digit Health. 2024 May;3(5):e0000516.

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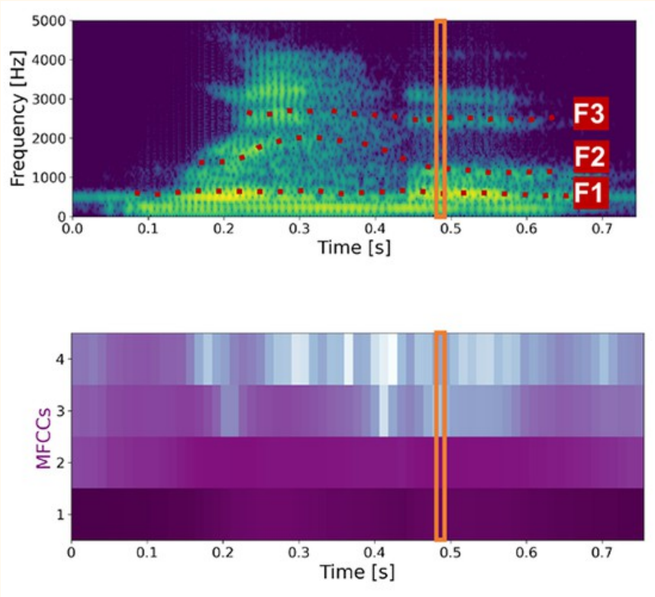






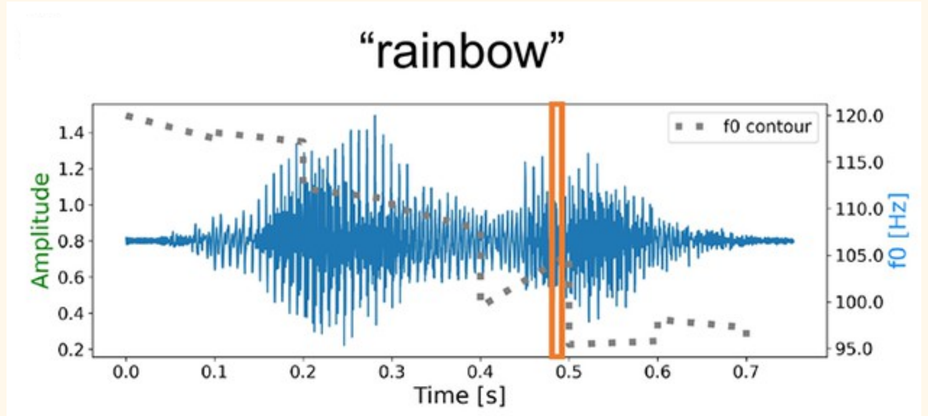
Questionnaire responses, dx status, demographics, confounders, etc.

## 1. Derived features




Computed stats: jitter, shimmer, spectral features, transcriptions, loudness, SI-SDR, etc.

## 2. Raw Audio




Low DM, Rao V, Randolph G, Song PC, Ghosh SS. Identifying bias in models that detect vocal fold paralysis from audio recordings using explainable machine learning and clinician ratings. PLOS Digit Health. 2024 May;3(5):e0000516.

B2AI VOICE

This repository is under review for potential modification in compliance with Administration directives.



Training opportunities for using the dataset: https://www.b2aivoicescholars.org/

Bridge2AI-Voice is a Precision Public Health grand challenge project funded by the NIH Common Fund Bridge2AI Program. Bridge2AI-Voice seeks to create a flagship, standardized, and ethically sourced dataset of 10,000 voices linked to health information to fuel research and discovery in voice biomarkers.

Our group aims to promote integration of voice as a biomarker of health in clinical care. To do so, we will generate a large multi-institutional, ethically sourced, and diverse voice dataset linked to multimodal health biomarkers to fuel voice AI research. Data collection is performed via a novel app (The Bridge2AI-Voice App) available as smartphone application linked to electronic health records (EHR). The app collected breathing sounds, voice, speech and linguistic tasks but also a considerable amount of health information through surveys and validated questionnaires. Other multimodal data collected includes imaging, genomics and respiratory function tests amongst others. The consortium is also addressing the growing ethical, legal, and social challenges surrounding voice AI, including risks of voice re-identification, vulnerabilities like voice AI hacking, concerns around voice data sharing and privacy, and the influence of gender and racial diversity on the development and application of these technologies.

Our developed best ethical practices through ethical inquiry have guided the development of voice collection protocol as well as data dissemination practices.

As voice is increasingly being recognized as a biomarker of health by the tech world and voice AI is gaining attention from multi-nationals such as Google, Amazon, Mozilla and Apple amongst others, many important issues related to patient privacy protection, ethical and fair representation of population, and clinical accuracy are arising. As a multidisciplinary group of academic experts, we aim to influence and guide the world of voice AI by ensuring patient protection through ethical and fairness principles and create safe innovative infrastructures to

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B2AI VOICE

# Data Access

## Access audio recordings

Instructions for accessing the data are provided on the data access page of the Bridge2AI learning module.

## Derived dataset

A derived dataset containing spectrograms and combined phenotypic data is available on PhysioNet under a permission access mechanism. Registration on PhysioNet and signing of a data use agreement will enable access. The latest version of the dataset is available at the following URL:

Bridge2AI-Voice: An ethically-sourced, diverse voice dataset linked to health information

## Dataset including raw audio

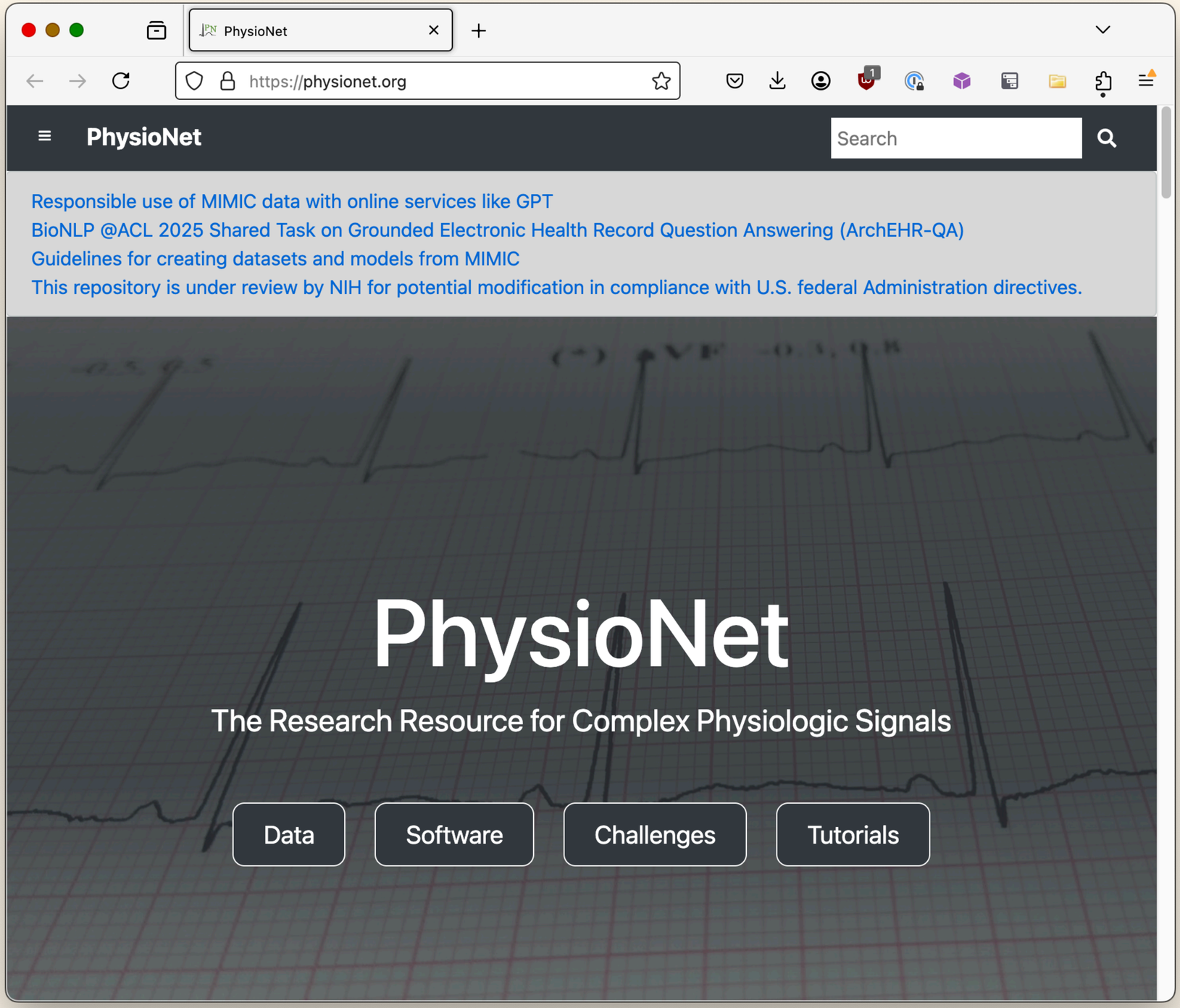
Raw audio is available under a controlled access mechanism. This process includes:

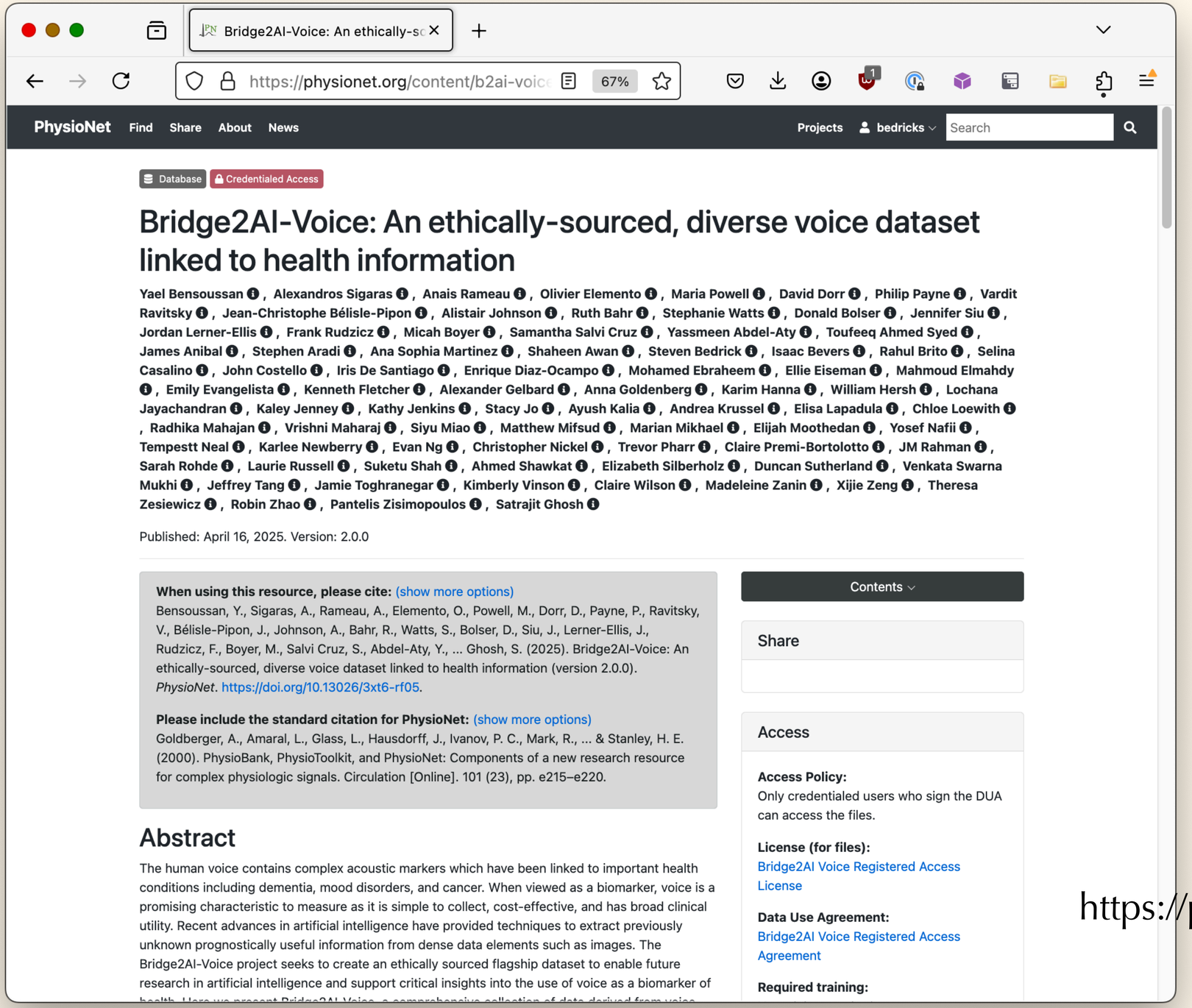
- Signing a Data Use Agreement (DUA): B2AI-Voice | Data Transfer and Use Agreement 2025.pdf
- Submitting a Data Access Request Form (DARF): B2AI-Voice | Data Access Request Form 2025.pdf

This process will require an institutional signature from a business official, a review by the Data Access Committee, and after approval, will be countersigned by USF. Once all forms are signed, the dataset can be obtained by submitting the counter-signed forms through PhysioNet at the following URL:

Bridge2AI-Voice: An ethically-sourced, diverse voice dataset linked to health information (Audio included)

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PhysioNet
https://physionet.org
PhysioNet
Search
Responsible use of MIMIC data with online services like GPT
BioNLP @ACL 2025 Shared Task on Grounded Electronic Health Record Question Answering (ArchEHR-QA)
Guidelines for creating datasets and models from MIMIC
This repository is under review by NIH for potential modification in compliance with U.S. federal Administration directives.
PhysioNet
The Research Resource for Complex Physiologic Signals
Data
Software
Challenges
Tutorials




https://physionet.org/content/b2ai-voice/

# Bridge2AI Voice Registered Access Agreement

**Authorized Researcher** means the researcher that signs this data access agreement, and is responsible for ensuring that the Research Team respects the conditions of this agreement in using the Data.

**Data** means the data, in whole or in part, that is made accessible to You by the Data Provider on the basis of this Agreement, including metadata, and any enriched or derived data. It includes, but is not limited to, representations, observations, narratives or measurements that could be used as the basis for further analysis, calculation or reasoning, regardless of the manner of their expression or the medium used to represent them.

**Data Provider** means the legal entity, or organisation, that grants Authorized Researchers access to the Data, and acts as the formal data custodian responsible for performing oversight and stewardship functions relative to the Data.

**Research Team** means You and other qualified researchers **at Your institution** that collaborate in using the Data for research.

**External Collaborators** means qualified researchers **outside of Your institution** that collaborate in Your research, and who have completed a Registered Data Access Agreement and are duly authorized to access and use the Data.

1. **Purpose.**

You agree to use the Data exclusively for commercial and non-commercial research purposes. In completing this agreement, You confirm that You and your Research Team have sufficient training and research expertise to use the data in a manner that protects the rights and interests of research participants. You agree to remain accountable for your use of the Data, as well as the use thereof by members of Your Research Team.

You agree to respect restrictions on data use arising from public law and private law, research ethics guidance, contracts, informed consent materials, consortium and institutional policies and research ethics requirements that are applicable to you.

You agree to obtain all approvals and authorizations that the Data Provider requires you to hold, and maintain them throughout the entire period of data use and data retention.

2. **Scientific Publications and Research Outputs.**

In publishing the findings from Your research, You agree to include the attribution statement that is associated to the dataset. Such an attribution statement provides clear instructions on how to attribute the Research Team, and/or the database or biobank of origin, and/or the dataset.

In addition, You agree that the publications arising from Your research will not contain the personal data of research participants. Such publications will not contain information that could reasonably be expected to cause a research participant to experience harm or stigmatization.

Posteriorgrams," ICASSP 2024 Workshop on Explainable Machine Learning for Speech and Audio, April 2024.

# Files

Total uncompressed size: 12.9 GB.

## Access the files

- Download the ZIP file (12.9 GB)
- Download the files using your terminal:

```
wget -r -N -c -np --user bedricks --ask-password https://physionet.org/files/b2ai-voice/3.0.0/
```

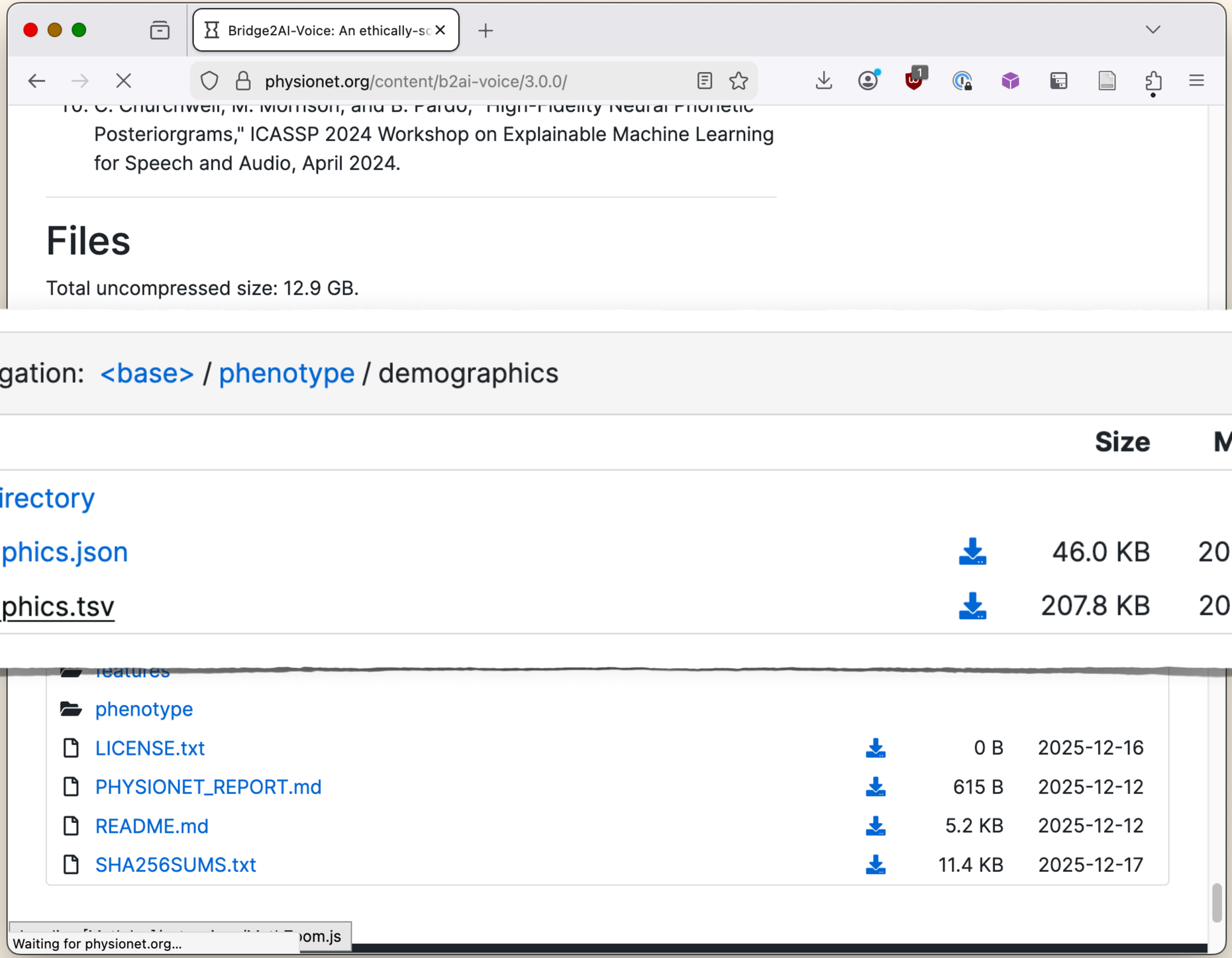
- Download the files using AWS command line tools:

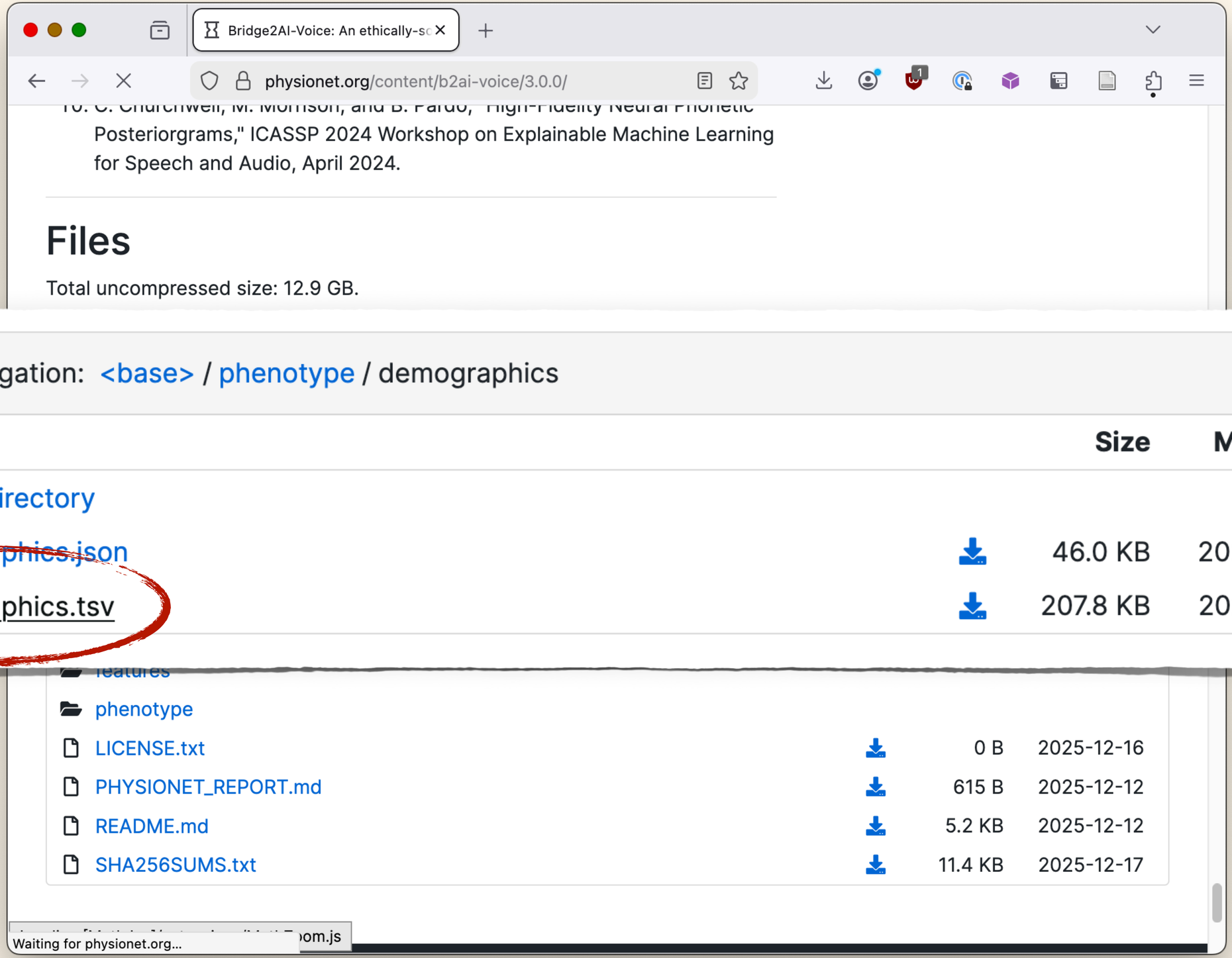
```
aws s3 sync s3://arn:aws:s3:us-east-1:724665945834:accesspoint/b2ai-voice-v3-0-0-01/b2ai-voice/3.0.0/
DESTINATION
```

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| phenotype | | |
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| PHYSIONET_REPORT.md | 615 B | 2025-12-12 |
| README.md | 5.2 KB | 2025-12-12 |
| SHA256SUMS.txt | 11.4 KB | 2025-12-17 |

Waiting for physionet.org...


Bridge2AI-Voice: An ethically-sc
physionet.org/content/b2ai-voice/3.0.0/
Posteriorgrams," ICASSP 2024 Workshop on Explainable Machine Learning for Speech and Audio, April 2024.
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Total uncompressed size: 12.9 GB.
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Size
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46.0 KB
2025-12-15
demographics.tsv
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2025-12-17
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Posteriorgrams," ICASSP 2024 Workshop on Explainable Machine Learning for Speech and Audio, April 2024.
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Bridge2AI-Voice: An ethically-s
polars.Series.to_numpy — Polar
nested numpy array to matrix -
https://physionet.org/content/b2ai-voic
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V., Sigaras, A., Elemento, O., & Bensoussan, Y. (2024). Bridge2AI-Voice: An ethically-sourced, diverse voice dataset linked to health information (version 1.0). Health Data Nexus. https://doi.org/10.57764/qb6h-em84

# Files

Total uncompressed size: 11.6 GB.

## Access the files

- Download the ZIP file (11.6 GB)
- Download the files using your terminal:

```
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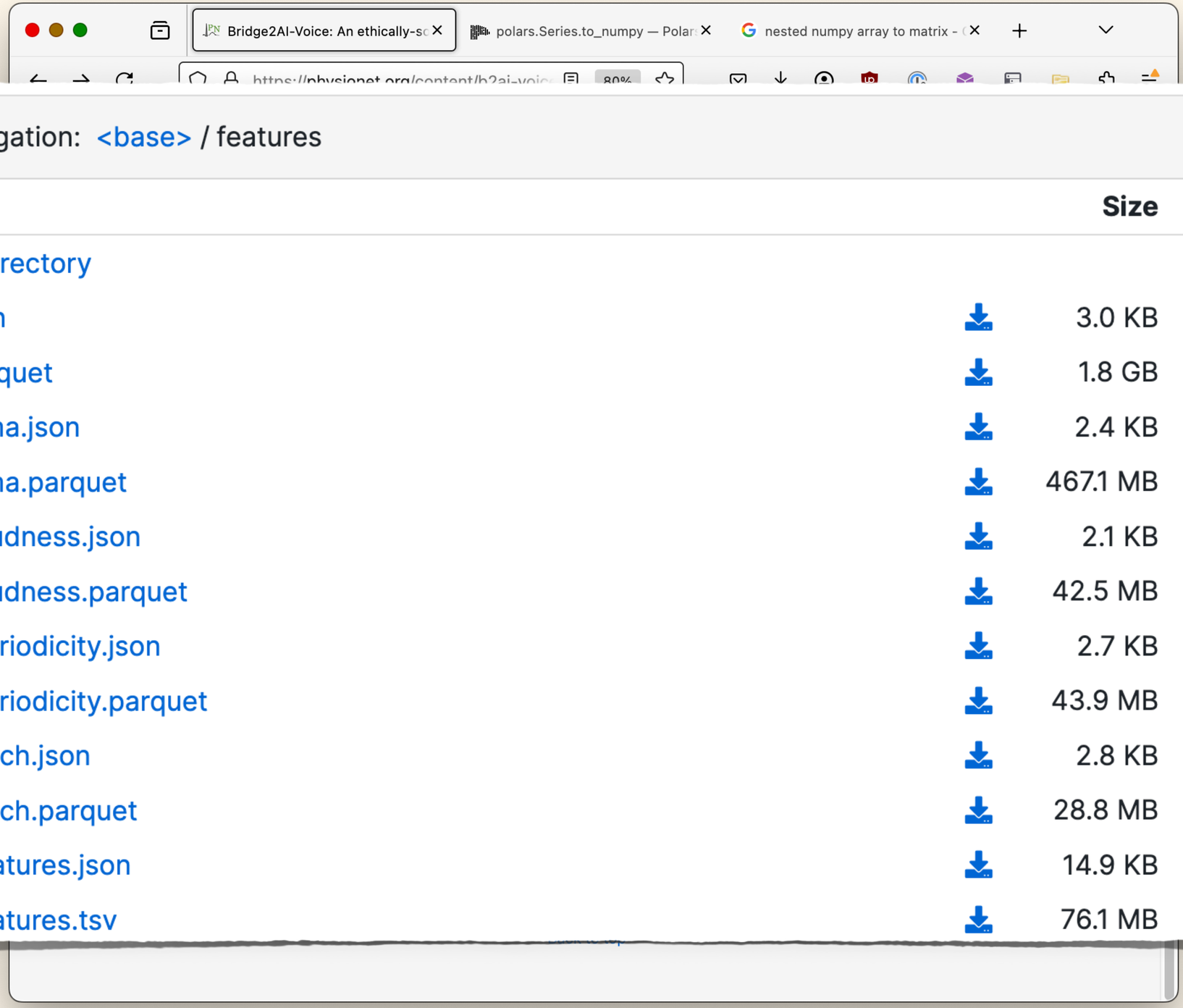
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| static_features.tsv | 45.7 MB | 2025-04-15 |

PhysioNet is a repository of freely-available medical research data, managed by the MIT Laboratory for Computational Physiology.

Supported by the National Institute of Biomedical Imaging and Bioengineering (NIBIB) under NIH grant number R01EB030362.

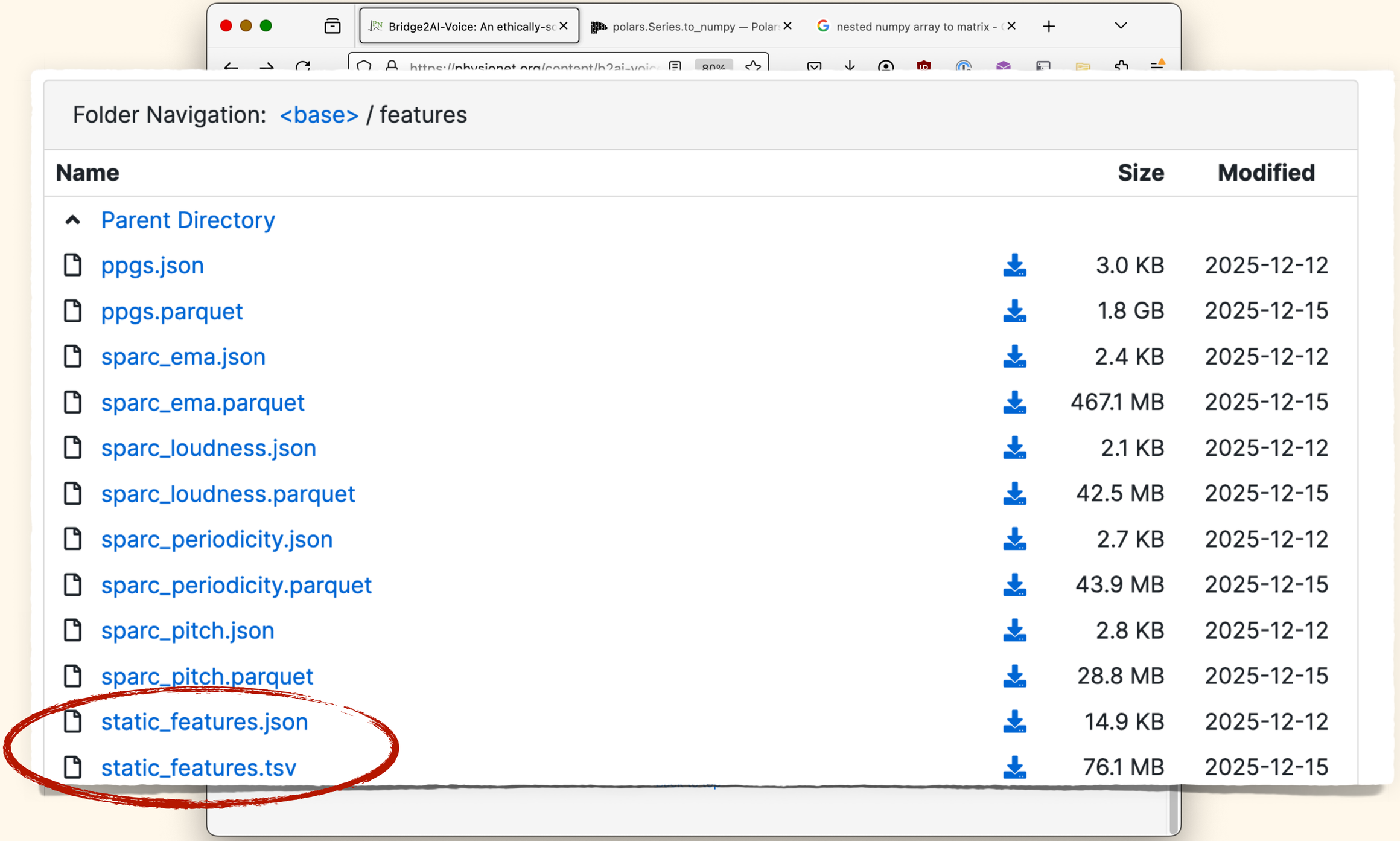
For more accessibility options, see the MIT Accessibility Page.

Back to top




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Bridge2AI-Voice: An ethically-s
polars.Series.to_numpy — Polar
nested numpy array to matrix -
80%
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static_features.tsv 76.1 MB 2025-12-15

# Important resources for working with this dataset:

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Primary documentation site: https://docs.b2ai-voice.org/

*Description of protocol, data use restrictions, study metadata, etc.*

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PhysioNet whitepaper: https://physionet.org/content/b2ai-voice/

*Shorter and more focused overview, including brief code example.*

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RedCap repository: https://github.com/eipm/bridge2ai-redcap

*Exhaustive description of all fields, examples of questionnaires, links to acoustic stimuli, map of protocols to questionnaires/acoustic tasks, etc.*

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FHIR schema: https://kind-lab.github.io/vbai-fhir/toc.html

*Machine-readable metadata schema, for those of us who are into that sort of thing.* 🤓

bridge2ai-redcap/README.m
github.com/eipm/bridge2ai-redcap/blob/main/README.md
New Chrome available






Voice as a Biomarker of Health

# bridge2ai-redcap

REDCap Data Dictionary and Metadata for the Bridge2AI project.




| Version | Date YYYY-mm-dd | Data Dictionary | Metadata | PDF's (English) | PDF's (Spanish) |
|---|---|---|---|---|---|
| v4.3.0 | 2025-10-24 | Data Dictionary | Metadata | Files | Files |

## 🤝 License

See LICENSE

## 📚 How to Cite

> Bensoussan, Y., Ghosh, S. S., Rameau, A., Boyer, M., Bahr, R., Watts, S., Rudzicz, F., Bolser, D., Lerner-Ellis, J., Awan, S., Powell, M. E., Belisle-Pipon, J.-C., Ravitsky, V., Johnson, A., Zisimopoulos, P., Tang, J., Sigaras, A., Elemento, O., Dorr, D., ... Bridge2AI-Voice. (2024). eipm/bridge2ai-redcap. Zenodo. https://zenodo.org/doi/10.5281/zenodo.12760724

## Task Overview

| # | Task Name | Type | Estimated Time to Complete (mm:ss) ⏳ | Protocol 📓 | Total # of Recordings 🔊 (if Acoustic Task) | Questionnaire Form PDF Link / Acoustic Task Description Youtube Link |
|---|---|---|---|---|---|---|
| 1 | Demographics | Questionnaire 📕 | 03:00 | Generic Protocol | - | PDF |
| 2 | Confounders | Questionnaire 📕 | 12:00 | Generic Protocol | - | PDF |
| 3 | Voice Perception | Questionnaire 📕 | 00:30 | Generic Protocol | - | PDF |
| 4 | VHI-10 | Questionnaire 📕 | 01:00 | Generic Protocol | - | PDF |
| 5 | Patient Health Questionnaire-9 | Questionnaire 📕 | 01:30 | Generic Protocol | - | PDF |
| 6 | GAD-7 Anxiety | Questionnaire 📕 | 01:30 | Generic Protocol | - | PDF |
| 7 | Prolonged Vowel | Acoustic Task 🎤 | 01:00 | Generic Protocol | 1 | YouTube Link |
| 8 | Maximum Phonation Time | Acoustic Task 🎤 | 02:00 | Generic Protocol | 3 | YouTube Link |

# Q Generic Confounders

Record ID

## Questionnaire - Metadata

Session ID

Questionnaire Started At

Questionnaire Completed At

Questionnaire Duration (seconds)

## Smoking

Have you smoked at least 100 cigarettes in your entire life? (There are 20 cigarettes in a pack.)?
- ◯ Yes
- ◯ No
- ◯ Don't know
- ◯ Prefer not to answer

Do you now smoke cigarettes every day, some days, or not at all?
- ◯ Every day
- ◯ Some days
- ◯ Not at all
- ◯ Don't know
- ◯ Prefer not to answer

How old were you when you first started regular cigarette smoking?
- ◯ Specify
- ◯ Don't know
- ◯ Prefer not to answer

Age:

If you have completely stopped smoking cigarettes, about how old were you when you stopped?
- ◯ Specify
- ◯ Don't know
- ◯ Prefer not to answer

Age when you stopped smoking:

How many years have you or did you smoke cigarettes?
- ◯ Specify
- ◯ Don't know
- ◯ Prefer not to answer

Number of years:





Native Hawaiian or other Pacific Islander
- ☐ Polynesian
- ☐ Micronesian
- ☐ Melanesian
- ☐ Other Pacific Islander

White
- ☐ European
- ☐ Middle Eastern Or North African
- ☐ Arab

Canadian Indigenous or Aboriginal
- ☐ First Nation
- ☐ Inuk/inuit
- ☐ Metis

If race not listed above, please specify:

What is your ethnic or cultural origin?
- ◯ Hispanic or Latino
- ◯ Not Hispanic or Latino
- ◯ Prefer not to answer

Check all that apply
- ☐ Spaniard
- ☐ Mexican
- ☐ Central American
- ☐ South American
- ☐ Latin American
- ☐ Puerto Rican
- ☐ Cuban
- ☐ Dominican

## Education

What is your highest level of education?
- ◯ No formal education
- ◯ Some elementary school
- ◯ Some secondary or high school education
- ◯ High School or secondary school degree complete
- ◯ Some college education
- ◯ Associate's or technical degree complete
- ◯ College or baccalaureate degree complete
- ◯ Some post-baccalaureate education
- ◯ Graduate or professional degree complete
- ◯ Doctoral or post graduate education
- ◯ Other
- ◯ Prefer not to answer

If you selected "other" level of education, please specify:

## Disability Questions

Are you deaf or do you have serious difficulty hearing?
- ◯ No
- ◯ Yes
- ◯ Prefer not to answer

Are you blind or do you have serious difficulty seeing, even when wearing glasses?
- ◯ No
- ◯ Yes
- ◯ Prefer not to answer

# Today's Plan

**Our dataset from 30,000 feet**

Getting hands dirty: problem formulation with our data

Practical demonstrations & exercises

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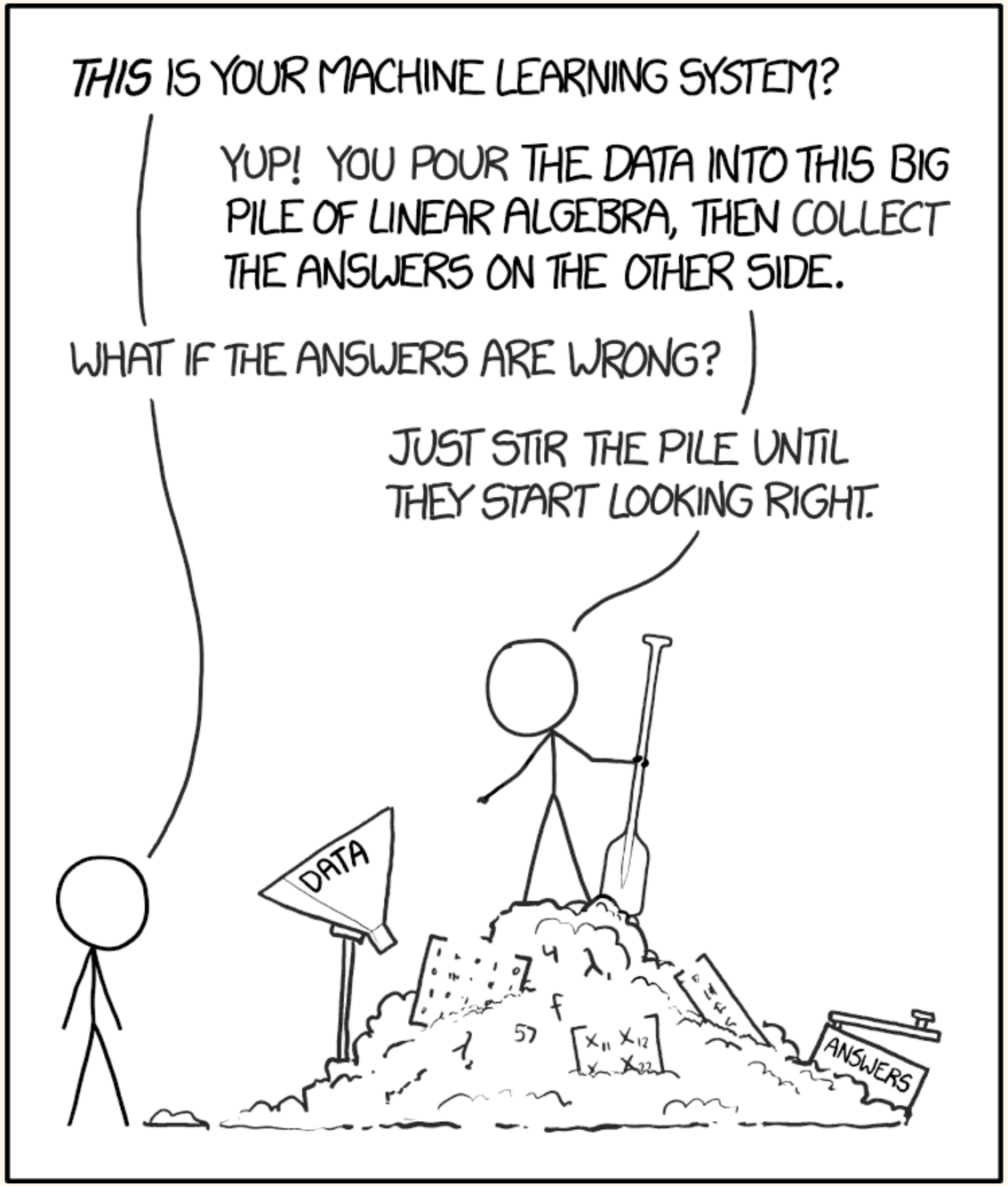
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*... and they all lived happily ever after...*




https://xkcd.com/1838/

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   *Alternatively: start with stimulus, find relevant protocols*

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*Inclusion criteria to consider:*

*Eligibility for protocol of interest*

*Case/control status*

*Sex at birth*

*Clinical confounders and covariates*

4. EDA to verify sample size, data completeness for stimuli, etc.

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6. Do whatever you were hoping to do vis-a-vis classification, regression, clustering, etc.

# Let's walk through an example!

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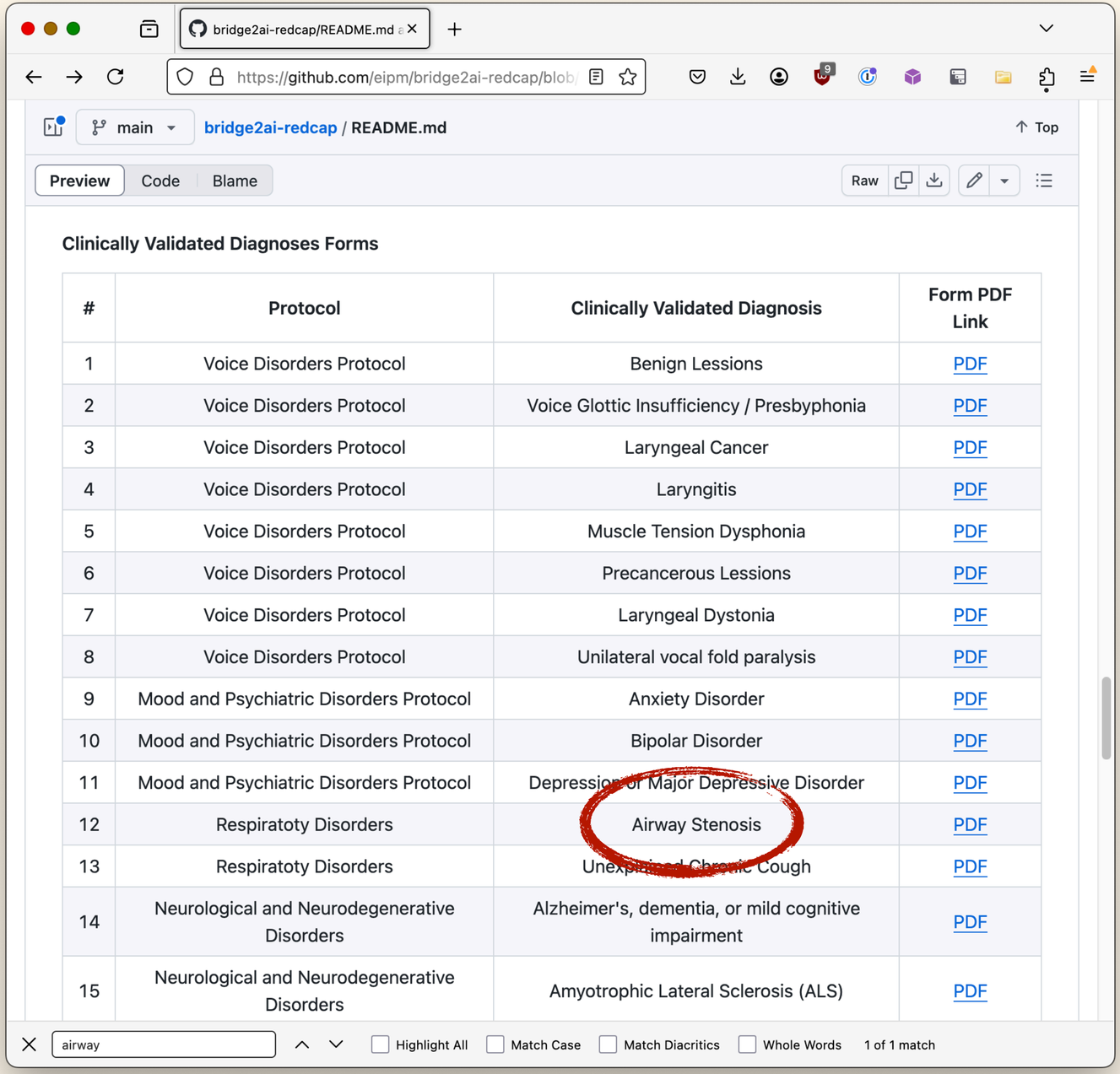
RQ: How do jitter and shimmer differ in patients with airway stenosis, as compared to controls?

1. Comparison of interest:

    *Airway stenosis vs. control population*

2. Determine protocols and stimuli:

    *Consult the documentation...*




## Clinically Validated Diagnoses Forms

| # | Protocol | Clinically Validated Diagnosis | Form PDF Link |
|---|---|---|---|
| 1 | Voice Disorders Protocol | Benign Lessions | PDF |
| 2 | Voice Disorders Protocol | Voice Glottic Insufficiency / Presbyphonia | PDF |
| 3 | Voice Disorders Protocol | Laryngeal Cancer | PDF |
| 4 | Voice Disorders Protocol | Laryngitis | PDF |
| 5 | Voice Disorders Protocol | Muscle Tension Dysphonia | PDF |
| 6 | Voice Disorders Protocol | Precancerous Lessions | PDF |
| 7 | Voice Disorders Protocol | Laryngeal Dystonia | PDF |
| 8 | Voice Disorders Protocol | Unilateral vocal fold paralysis | PDF |
| 9 | Mood and Psychiatric Disorders Protocol | Anxiety Disorder | PDF |
| 10 | Mood and Psychiatric Disorders Protocol | Bipolar Disorder | PDF |
| 11 | Mood and Psychiatric Disorders Protocol | Depression or Major Depressive Disorder | PDF |
| 12 | Respiratoty Disorders | Airway Stenosis | PDF |
| 13 | Respiratoty Disorders | Unexplained Chronic Cough | PDF |
| 14 | Neurological and Neurodegenerative Disorders | Alzheimer's, dementia, or mild cognitive impairment | PDF |
| 15 | Neurological and Neurodegenerative Disorders | Amyotrophic Lateral Sclerosis (ALS) | PDF |

*Confidential*



# D Resp Airway Stenosis

Record ID _____

### 1. Gold Standard Diagnosis

Has the patient had pulmonary function testing within the past 3 months?
- ◯ No
- ◯ Yes
- ◯ Not certain

Please upload the PFT document

Does the patient have airway stenosis confirmed by a laryngoscopy, bronchoscopy or CT imaging?
- ◯ No
- ◯ Yes
- ◯ Not certain

Confirmed by
- ☐ Confirmed by bronchoscopy or laryngoscopy
- ☐ Confirmed by CT imaging

### 2. Disease Subtype

Disease Subtype
- ◯ Subglottic Stenosis
- ◯ Bilateral Vocal fold immobility or Glottic Stenosis
- ◯ Supraglottic Stenosis
- ◯ Tracheal Stenosis
- ◯ Multi-Level Upper Airway Stenosis

Bilateral Vocal fold immobility or Glottic Stenosis Subtype
- ◯ Posterior Glottic Stenosis
- ◯ Bilateral Vocal Fold Paralysis
- ◯ Anterior glottic Web
- ◯ Not certain or TBD
- ◯ Other

Other, Please specify _____

### 3. Etiology

What is the etiology?
- ◯ Not certain or TBD
- ◯ Iatrogenic (surgery or intubation)
- ◯ Idiopathic
- ◯ Auto-immune or granulomatous disease
- ◯ Radiation
- ◯ GERD
- ◯ Caustic Injury or Inhalation Injury
- ◯ Infectious (Diphtheria, TB, Bacterial Tracheitis, other)
- ◯ Other

Other, Please specify _____

# Let's walk through an example!

2. Determine protocols and stimuli:

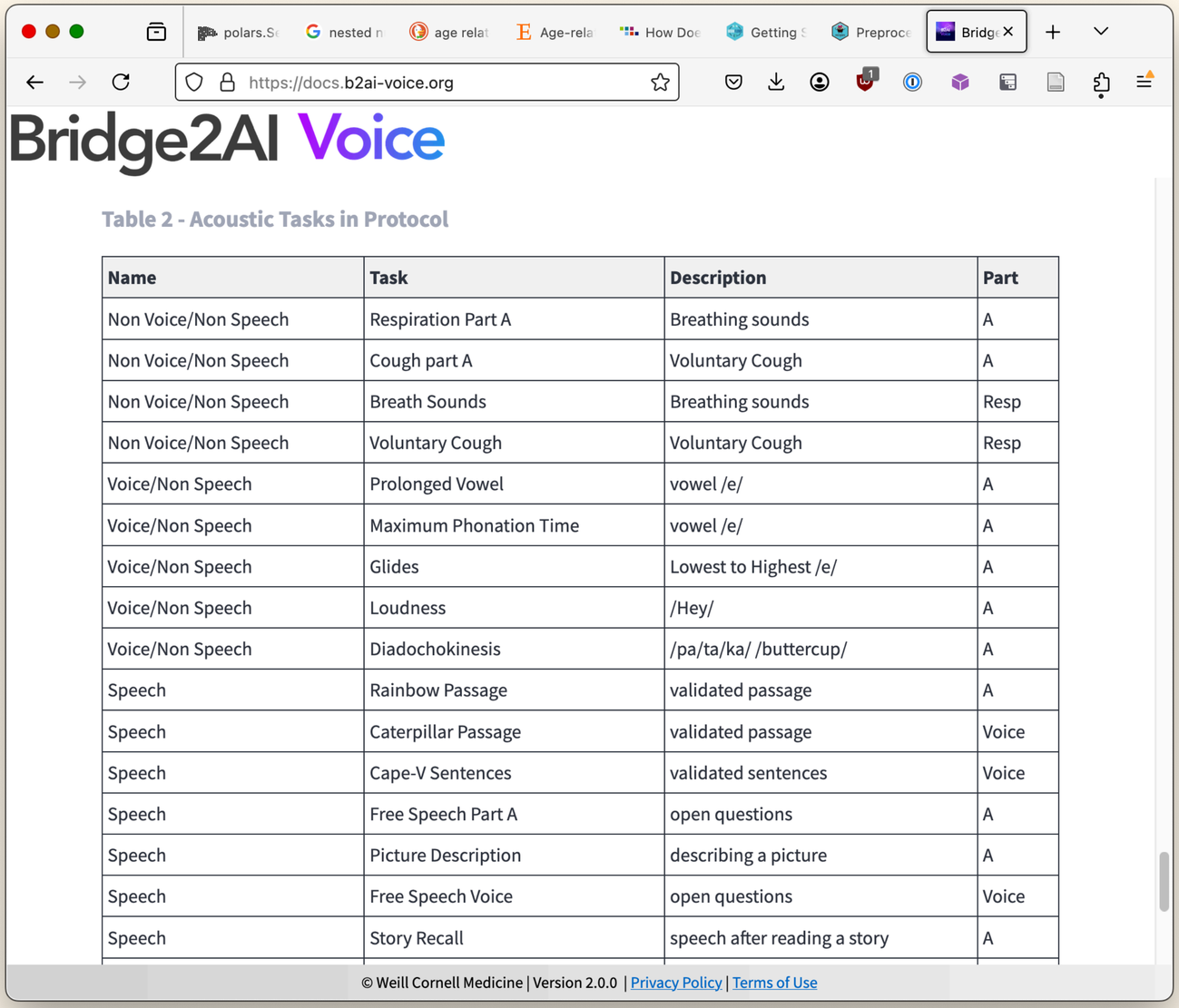


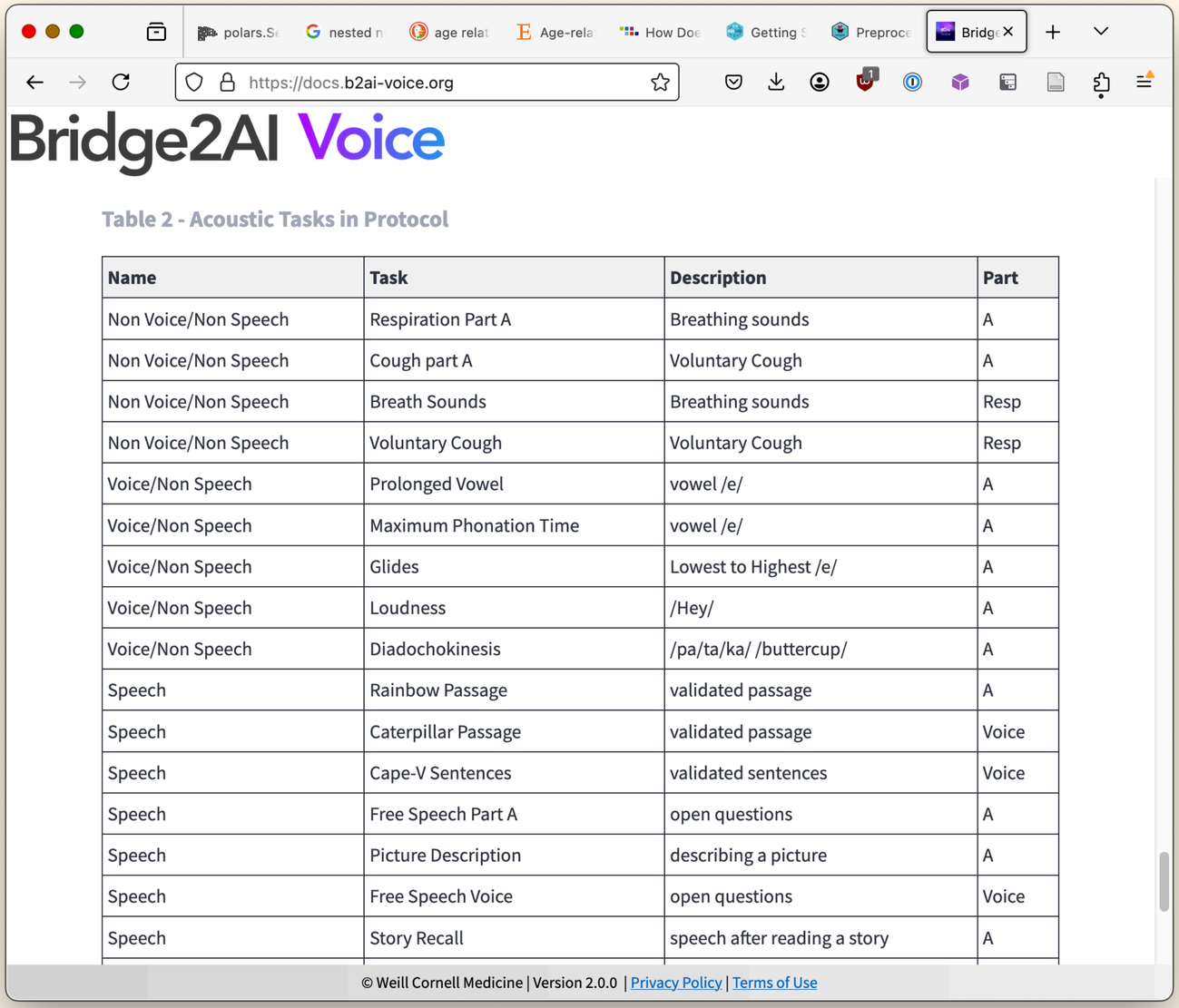

Table 2 - Acoustic Tasks in Protocol

| Name | Task | Description | Part |
|---|---|---|---|
| Non Voice/Non Speech | Respiration Part A | Breathing sounds | A |
| Non Voice/Non Speech | Cough part A | Voluntary Cough | A |
| Non Voice/Non Speech | Breath Sounds | Breathing sounds | Resp |
| Non Voice/Non Speech | Voluntary Cough | Voluntary Cough | Resp |
| Voice/Non Speech | Prolonged Vowel | vowel /e/ | A |
| Voice/Non Speech | Maximum Phonation Time | vowel /e/ | A |
| Voice/Non Speech | Glides | Lowest to Highest /e/ | A |
| Voice/Non Speech | Loudness | /Hey/ | A |
| Voice/Non Speech | Diadochokinesis | /pa/ta/ka/ /buttercup/ | A |
| Speech | Rainbow Passage | validated passage | A |
| Speech | Caterpillar Passage | validated passage | Voice |
| Speech | Cape-V Sentences | validated sentences | Voice |
| Speech | Free Speech Part A | open questions | A |
| Speech | Picture Description | describing a picture | A |
| Speech | Free Speech Voice | open questions | Voice |
| Speech | Story Recall | speech after reading a story | A |

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2. Determine protocols and stimuli:

*Airway stenosis cases, + controls from resp. protocol*



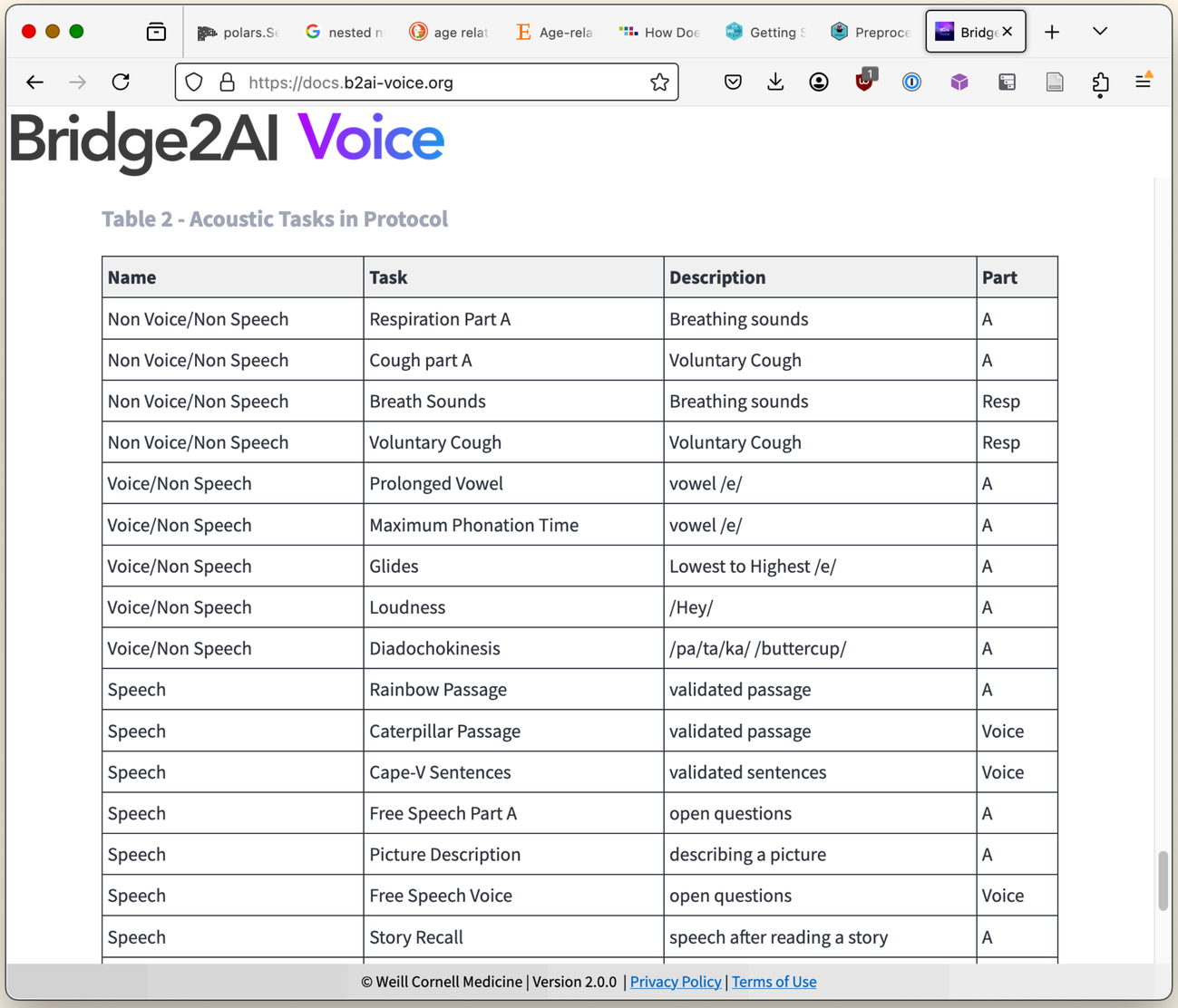

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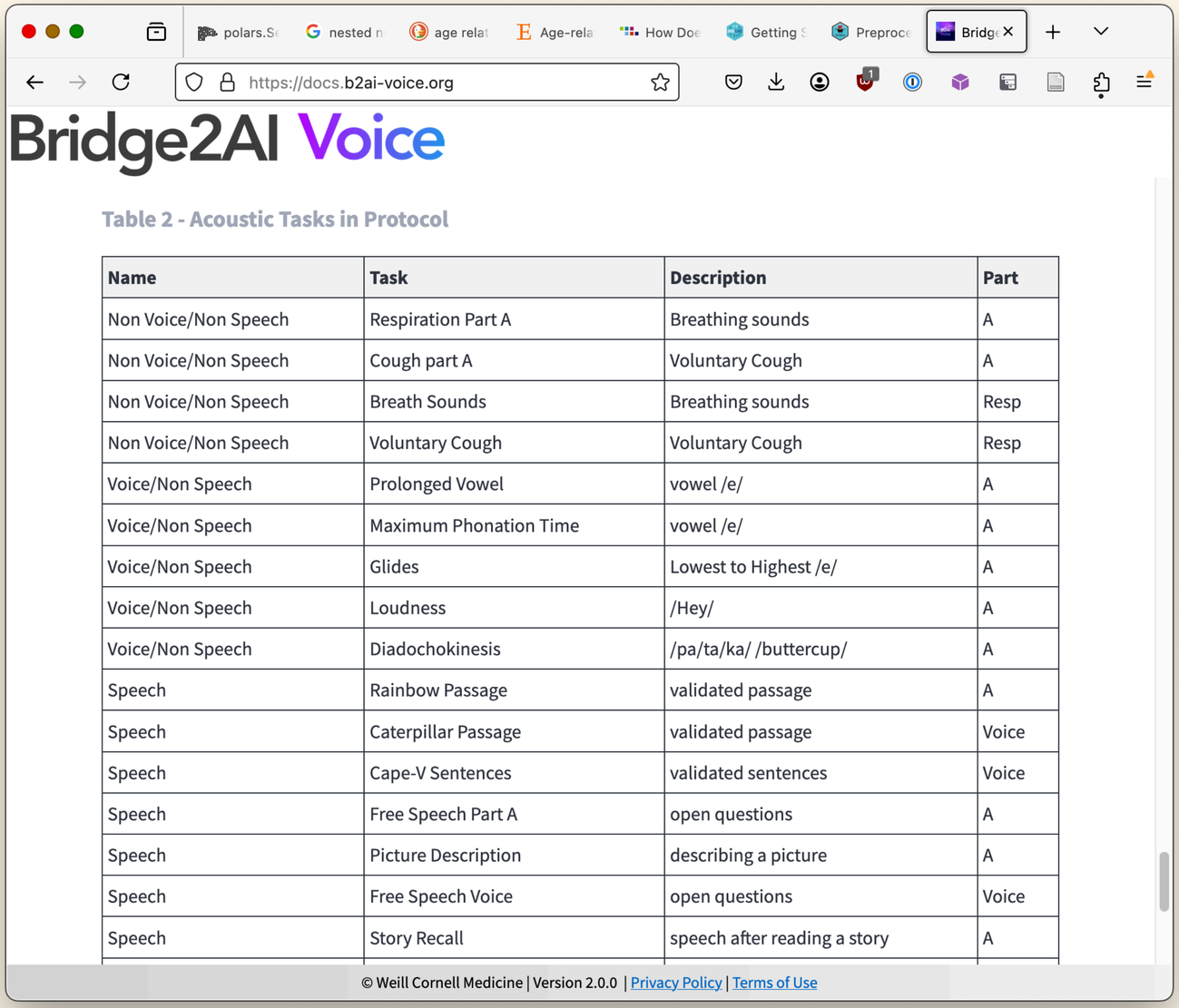
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*Let's hop over to R...*

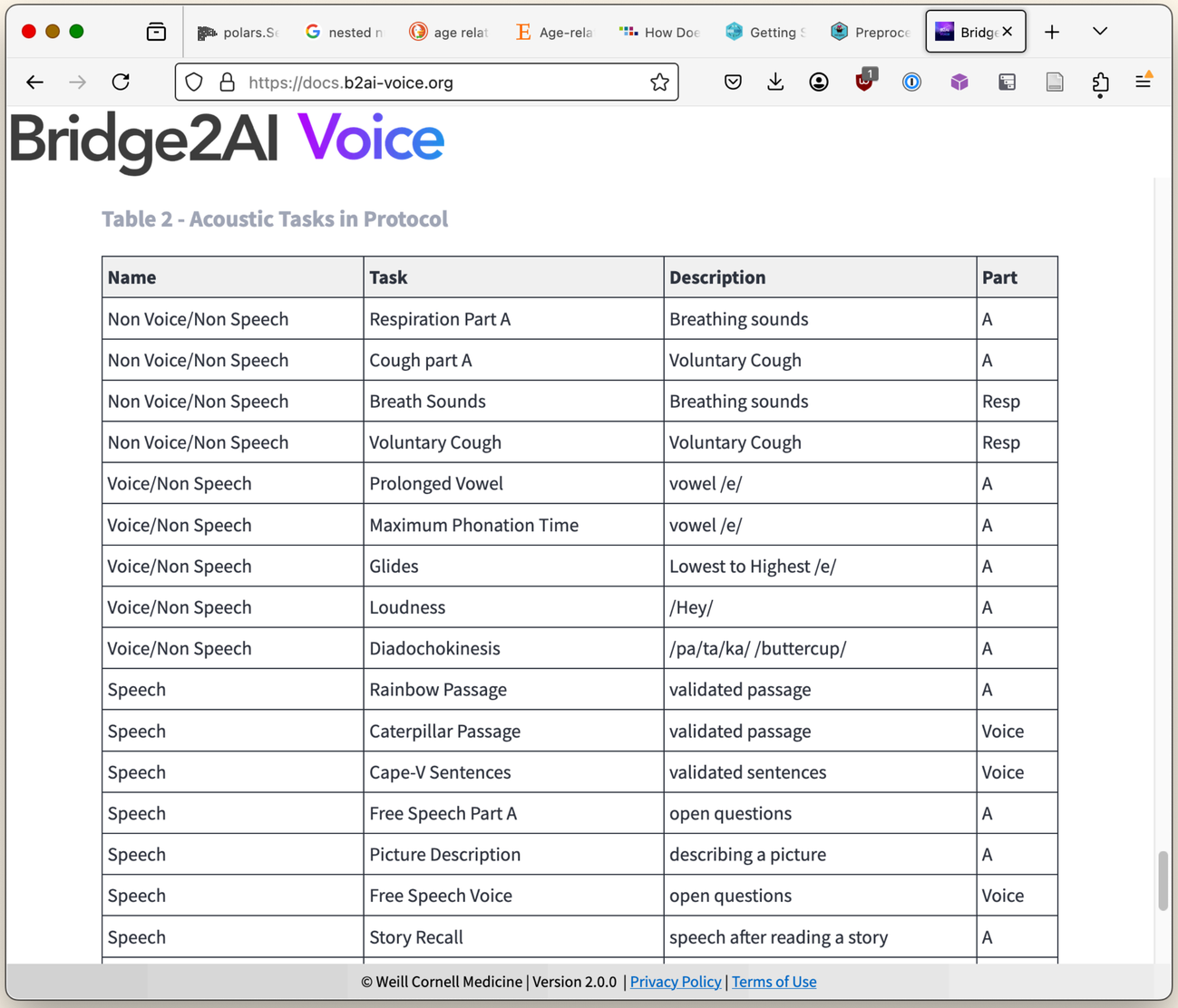



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https://dmice.ohsu.edu/bedricks/b2ai_hackathon_may_2026/

# Thank you!

# Questions?

bedricks@ohsu.edu

**A huge "thank you" to:**

The *entire* Bridge2AI-Voice team, across all of our sites

Our participants, for donating their data and time

The rest of the Bridge2AI Consortium

**The NIH! We love you guys!** 3OT2OD032720

